CS Fundamentals Revision Handbook

Prepo by Avneesh — OOP, DBMS, SQL, OS & Networks | Category: QUICK REVISION NOTES

1. OOP Principles

• Encapsulation: Binding data & methods inside class; protection through private/protected access.
• Abstraction: Hiding implementation complexity and exposing interfaces (Interfaces & Abstract Classes).
• Inheritance: Code reuse through IS-A hierarchy; Single, Multilevel, Multiple (via interfaces).
• Polymorphism: Compile-time (Method Overloading) vs Runtime (Virtual Method Overriding).

2. DBMS & SQL Core

• ACID: Atomicity (All or none), Consistency (State valid), Isolation (Concurrent safe), Durability (Persisted).
• Normalization: 1NF (Atomic fields), 2NF (No partial dependency), 3NF (No transitive dependency), BCNF.
• Joins: INNER (matching rows), LEFT/RIGHT (preserve unmatching side), CROSS (Cartesian product).
• GROUP BY & HAVING: GROUP BY aggregates data; HAVING filters aggregate metrics (WHERE cannot).

3. Operating Systems & Computer Networks

• Deadlock Coffman Conditions: Mutual Exclusion, Hold and Wait, No Preemption, Circular Wait.
• Process vs Thread: Processes have independent memory; threads share heap, code, data of parent.
• OSI Layers: Physical, Data Link, Network (IP), Transport (TCP/UDP), Session, Presentation, Application.
• TCP vs UDP: TCP is connection-oriented, reliable (3-way handshake); UDP is connectionless and fast.